— ECAF 2026

What if the **AI** you used to **apply** is why you did not get **hired**?

When LLMs Evaluate LLMs in Hiring: A Multi-Model Auditing Framework for Generative Preference Bias in AI-Assisted Recruitment

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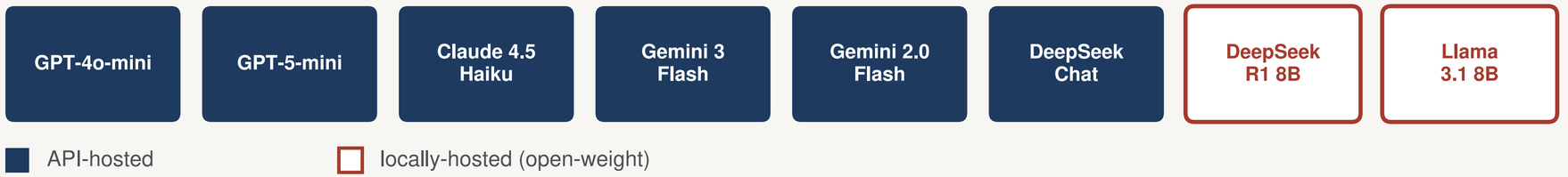
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— THE AUDIT

PHASE 1

Generation

8 writer models draft cover letters (350–400 words, gender-neutral persona)
for 50 CVs × 10 jobs — stratified top-25 high-fit / next-25 moderate-fit by cosine similarity.



PHASE 2

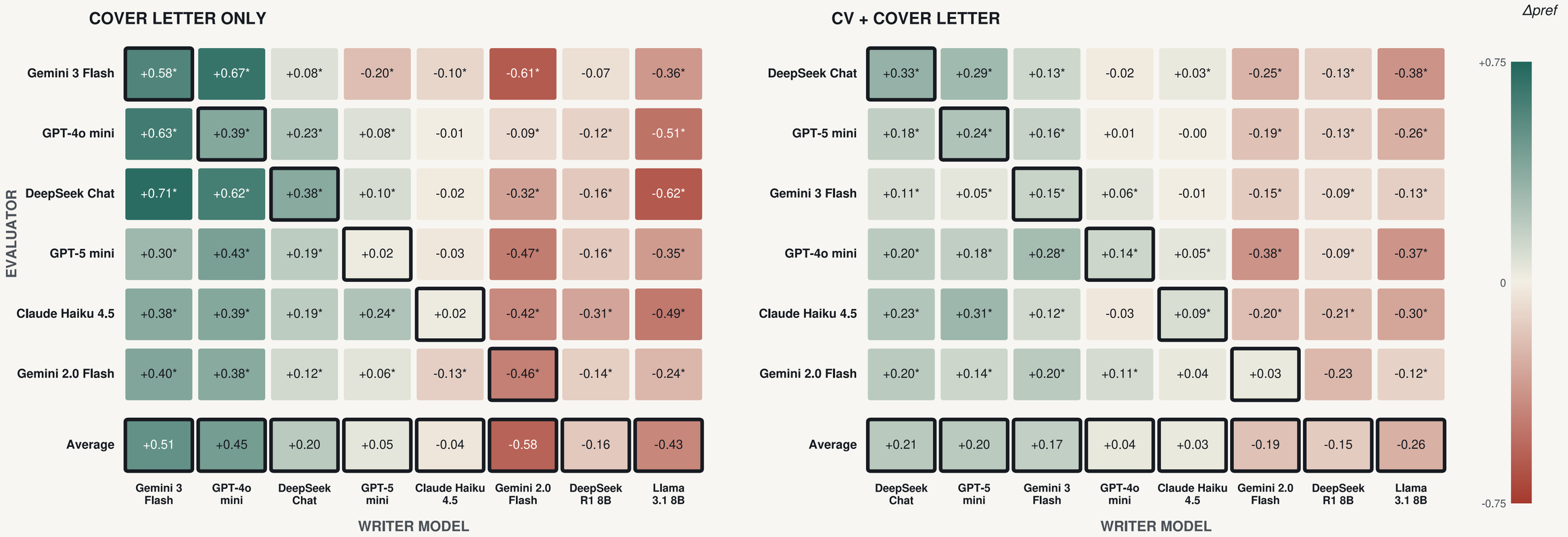
Evaluation

6 API-hosted evaluator models score every candidate, × 4 iterations each, under:



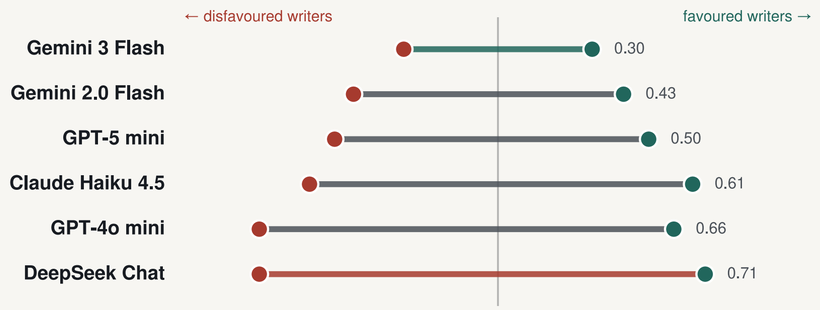
THE FINDINGS

A Generative Preference Bias

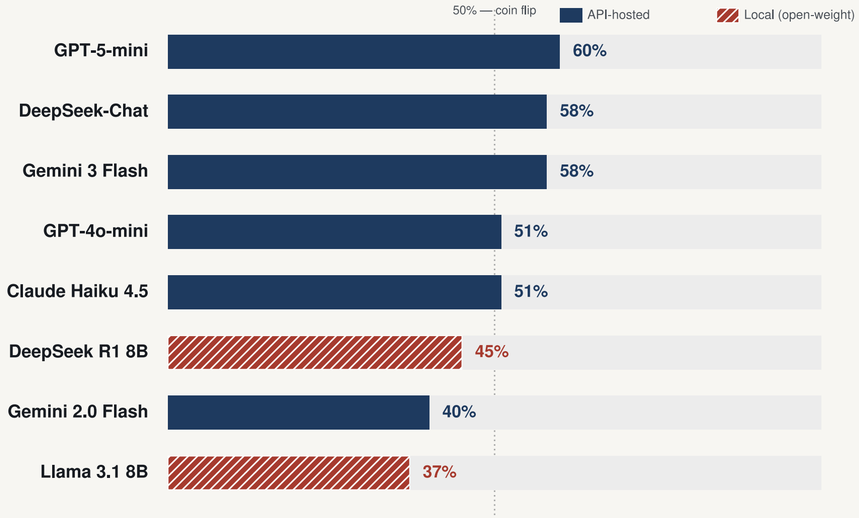


— THE FINDINGS

B Evaluator Fairness



C Head-to-head Win Rate



D Rank Displacement

25 equally-qualified (high-fit) candidates. Worst case: high-fit group writes with Llama-3.1-8B, moderate-fit group writes with a top proprietary model.



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qualified candidates displaced purely by model mismatch.

An Inequity Built on Tool Access

FOR APPLICANTS

Tool choice carries real, unrecoverable consequences. Skipping AI is worse.

FOR EMPLOYERS

Fairer models exist and are identifiable. Screening-model choice is a design decision.

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